Architectural Decision Record (ADR)

Title: Adopt a Hybrid Tech Stack for Cross-Platform AI Deployment in Resource-Constrained Environments
Status: Approved
Context:
The project requires deploying AI models (e.g., computer vision for agriculture/health) on **mobile devices** and **augmented reality (AR) glasses** with strict constraints:

- **Offline operation** (no cloud dependency).
- **Computational efficiency** (support for entry-level hardware: ≤2 GB RAM, limited GPU/CPU).
- **Cross-platform compatibility** (iOS, Android, AR glasses like HoloLens/Magic Leap).
- **Real-time performance** for tasks like object detection in agriculture or AR overlays.

Decision

Adopt a **hybrid technology stack** combining:

1. **Python/PyTorch** for model development and training.
2. **ONNX/TensorFlow Lite** for model optimization and deployment.
3. **Flutter** for cross-platform mobile apps.
4. **Unity + TensorFlow Lite/ONNX** for AR glasses integration.
5. **Platform-specific SDKs** (e.g., HoloLens/Magic Leap) for AR hardware.

Drivers

1. **Efficiency & Portability**:
    - Lightweight models (e.g., YOLOv8nano, EfficientNetB0) require frameworks like **ONNX Runtime** and **TensorFlow Lite** for quantization and edge optimization.
2. **Cross-Platform Support**:
    - **Flutter** enables code reuse for iOS/Android, reducing development effort.
    - **Unity** is industry-standard for AR/3D visualization.
3. **Offline & Real-Time Needs**:
    - ONNX/TFLite enable on-device inference without cloud connectivity.
4. **Research-Friendly Development**:
    - **PyTorch** offers flexibility for custom model architectures and training.

Alternatives Considered

1. **Native-Only Approach**:
   - Use **Core ML** (iOS) and **MLKit** (Android) for mobile, **C#** for AR glasses.
   - *Rejected*: Limits cross-platform code reuse and complicates model optimization.
2. **Single Framework for All**:
   - Rely solely on **TensorFlow Lite** or **ONNX** for all deployment.
   - *Rejected*: Lacks native AR/3D integration (Unity/ARKit) and poor mobile UX compared to Flutter.
3. **No AR Support**:
   - Focus only on mobile.
   - *Rejected*: Excludes critical use cases (e.g., AR overlays for pest detection in crops).
4. **Heavy Models with Cloud Offload**:
   - Use larger models with cloud inference.
   - *Rejected*: Violates offline requirement and introduces latency.

Rationale

Choice	Pros	Cons
PyTorch + ONNX/TFLite	- PyTorch's flexibility for research; ONNX/TFLite enable cross-platform optimization.	- Learning curve for PyTorch; model conversion adds complexity.
Flutter	- Single-codebase for iOS/Android; rich widget ecosystem.	- Performance overhead vs. native Swift/Kotlin (mitigated by TFLite).
Unity + AR SDKs	- Robust 3D/AR tooling; integrates with TFLite/ONNX for inference.	- Steeper learning curve for Unity; larger app size.
Platform-Specific SDKs	Direct hardware access for AR glasses (e.g., HoloLens tracking).	- Fragmented codebase; increases maintenance effort.

Final Justification:
The hybrid stack balances **efficiency**, **cross-platform reach**, and **performance**. PyTorch accelerates model development, while ONNX/TFLite ensure lightweight deployment. Flutter/Unity unify mobile/AR experiences, and platform-specific SDKs address hardware constraints. This approach meets all technical criteria while minimizing lock-in.

Implications

1. **Complexity**: Managing multiple frameworks (PyTorch, Flutter, Unity) increases initial development effort.
2. **Performance Tuning**: Requires iterative profiling (e.g., TensorFlow Profiler, AR emulator testing).
3. **Future Compatibility**: AR glasses platforms (e.g., HoloLens 2 vs. Magic Leap 2) may require SDK updates.

References

1. **Model Optimization**:
   - TensorFlow Lite Model Optimization Guide
   - ONNX Runtime Performance Best Practices
2. **Cross-Platform Development**:
   - Flutter Mobile SDK
   - Unity TensorFlow Lite Plugin
3. **AR Glasses Integration**:
   - HoloLens SDK
   - Magic Leap SDK
4. **Model Selection Justification**:
   - YOLOv8nano for Edge Devices (Ultra-lightweight, <2MB).
   - EfficientNetB0 for Agriculture (High accuracy with low compute).
5. **Project Constraints**:
   - Allen Institute Model Weights (Open-source foundation models).

Reviewers: AI Team, Mobile/AR Engineers
Next Review Date: [Insert Date]